Warm Up (everyone): Find $\frac{d^2y}{dx^2}$ if $x^4+y^4=16$.

1.) If $f(x)+x^2(f(x))^3=10$ and $f(1)=2$, find $f'(1)$.

2.) Use implicit differentiation to find an equation of the tangent line to the curve $y\sin(2x)=x\cos(2y)$ at the point $(\frac{\pi}{2},\frac{\pi}{4})$.

3.) Show that the sum of the x- and y-intercepts of any tangent line to the curve $\sqrt{x}+\sqrt{y}=\sqrt{c}$ is equal to c.

4.) (a) Two curves are said to be *orthogonal* if their tangent lines are perpendicular at each point of intersection. Show that the curves

$$x^2+y^2=ax \quad \text{and} \quad x^2+y^2=by$$

are *orthogonal trajectories* of each other; that is, ever curve in one family is orthogonal to every curve in the other family. Sketch both families of curves on the same axes.

(b) Find the value of the number a such that the families of curves

$$y=(x+c)^{-1} \quad \text{and} \quad y=a(x+k)^{1/3}$$

are orthogonal trajectories.

5.) (a) The *van der Waals* equation for n moles of a gas is

$$\left(P+\frac{n^2a}{V^2}\right)(V-nb)=nRT$$

where P is the pressure, V is the volume, and T is the temperature of the gas. The constant R is the universal gas constant and a and b are positive constants that are characteristic of a particular gas. If T remains constant, use implicit differentiation to find $\frac{dV}{dP}$.

(b) Find the rate of change of volume with respect to pressure of 1 mole of CO_2 at a volume of $V=10$L and a pressure of $P=2.5$atm. Use $a=3.592\text{L}^2\text{-atm/mole}^2$ and $b=0.04267\text{L/mole}$.